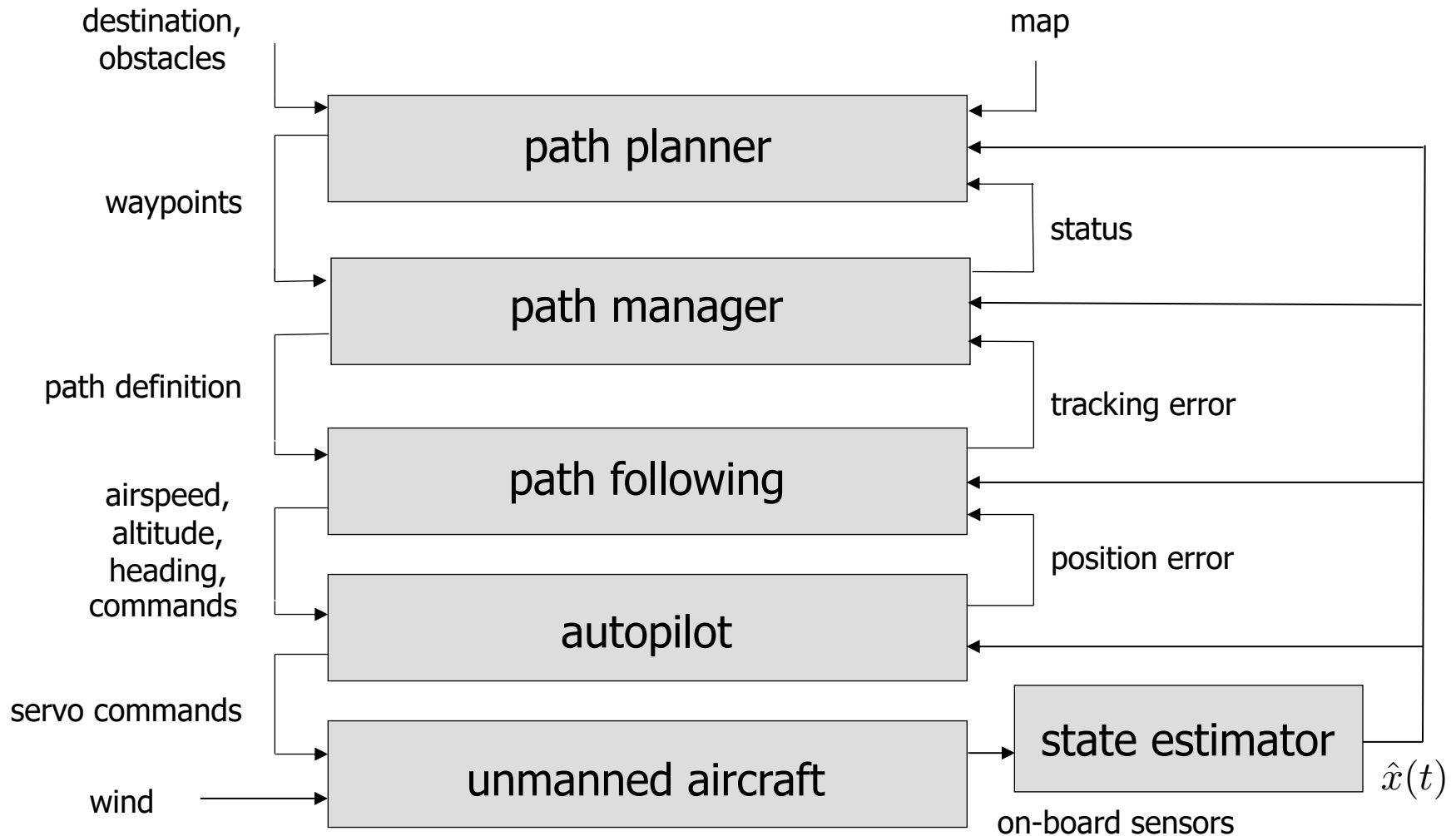


A yellow and black glider is flying in the upper left portion of the frame. The background features a vast, cloudy sky with patches of blue. In the lower third of the image, a farm landscape is visible, including a white barn, a windmill, and a line of trees, all situated behind a field of tall, golden-brown crops.

Chapter 10

Path Following

Control Architecture



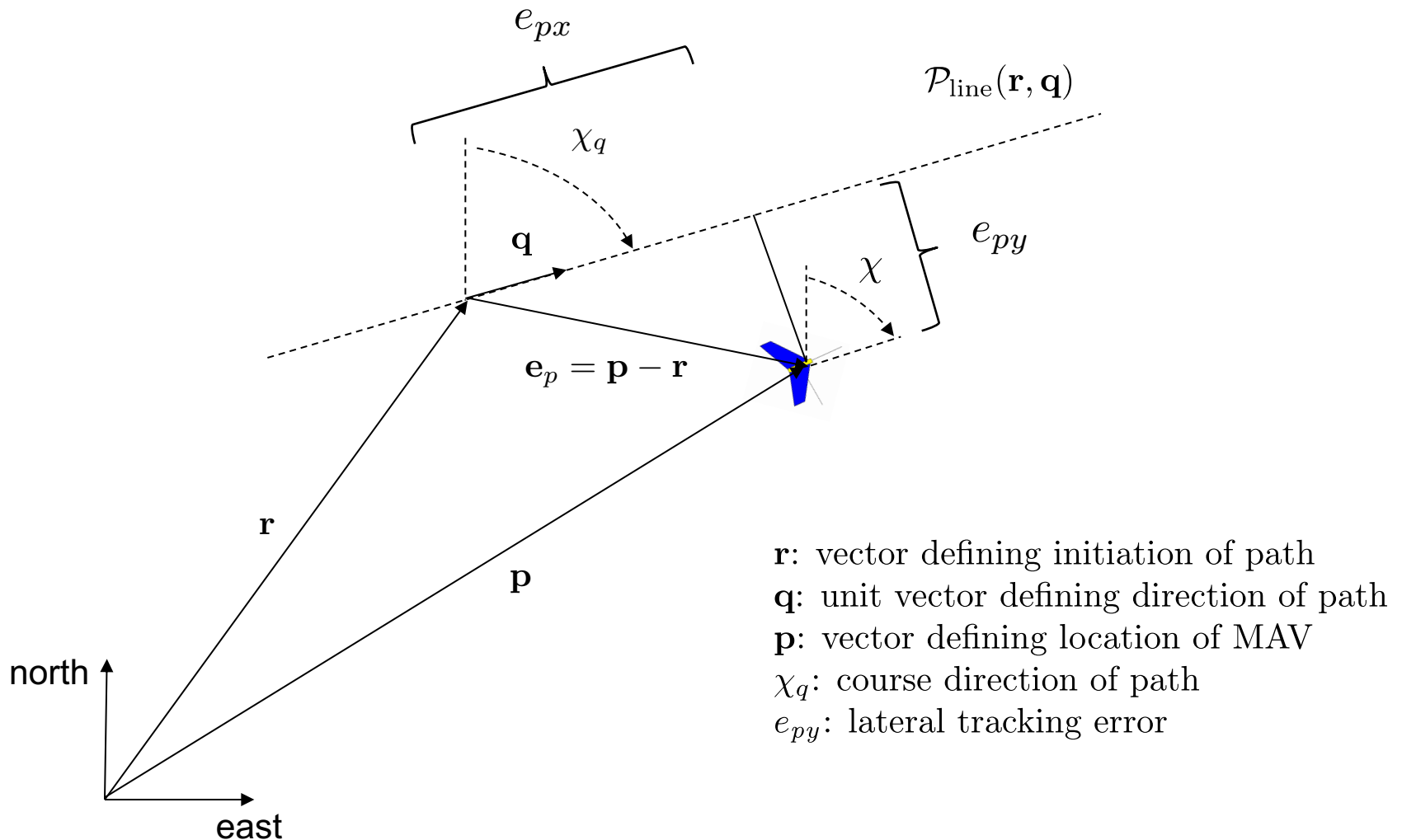
Path Following

- For small UAVs, a major issue is wind
 - Always present to some degree
 - Usually significant with respect to commanded airspeed
- Wind makes traditional trajectory tracking approaches difficult, if not infeasible
 - Have to know the wind precisely at every instant to determine desired airspeed
- Better approach: path following
- Rather than “follow this trajectory”, we control UAV to “stay on this path”

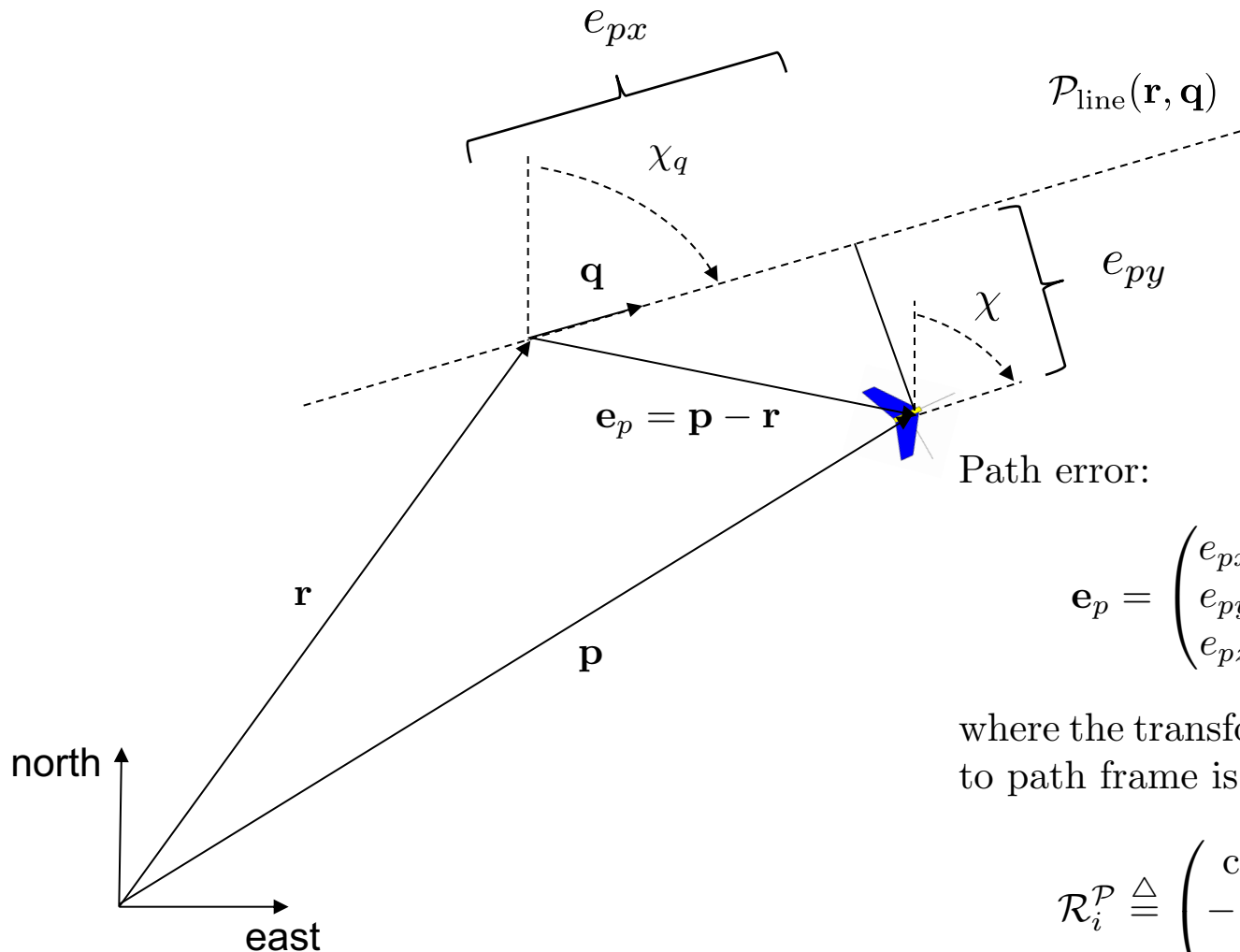
Path Types

- We will focus on two types of paths to follow:
 - Straight lines between two points in 3-D
 - Inclination of path within climb capabilities of UAV
 - Circular orbits or arcs in the horizontal plane
- Paths for common applications can be built up from these path primitives
 - Methods for following other types of paths found in literature

Straight Line Path Description



Lateral Tracking Problem



$$\mathbf{e}_p = \begin{pmatrix} e_{px} \\ e_{py} \\ e_{pz} \end{pmatrix} \triangleq \mathcal{R}_i^{\mathcal{P}} (\mathbf{p}^i - \mathbf{r}^i)$$

where the transformation from inertial frame to path frame is

$$\mathcal{R}_i^{\mathcal{P}} \triangleq \begin{pmatrix} \cos \chi_q & \sin \chi_q & 0 \\ -\sin \chi_q & \cos \chi_q & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Lateral Tracking Problem

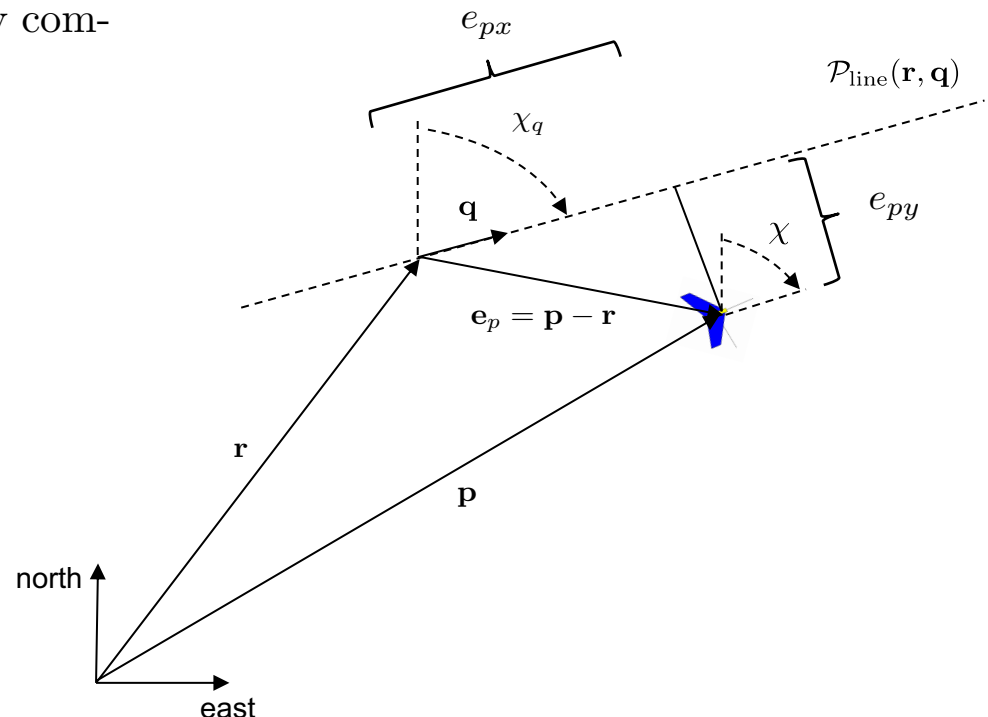
Relative error dynamics in path frame:

$$\begin{aligned} \begin{pmatrix} \dot{e}_{px} \\ \dot{e}_{py} \end{pmatrix} &= \begin{pmatrix} \cos \chi_q & \sin \chi_q \\ -\sin \chi_q & \cos \chi_q \end{pmatrix} \begin{pmatrix} V_g \cos \chi \\ V_g \sin \chi \end{pmatrix} \\ &= V_g \begin{pmatrix} \cos(\chi - \chi_q) \\ \sin(\chi - \chi_q) \end{pmatrix} \end{aligned}$$

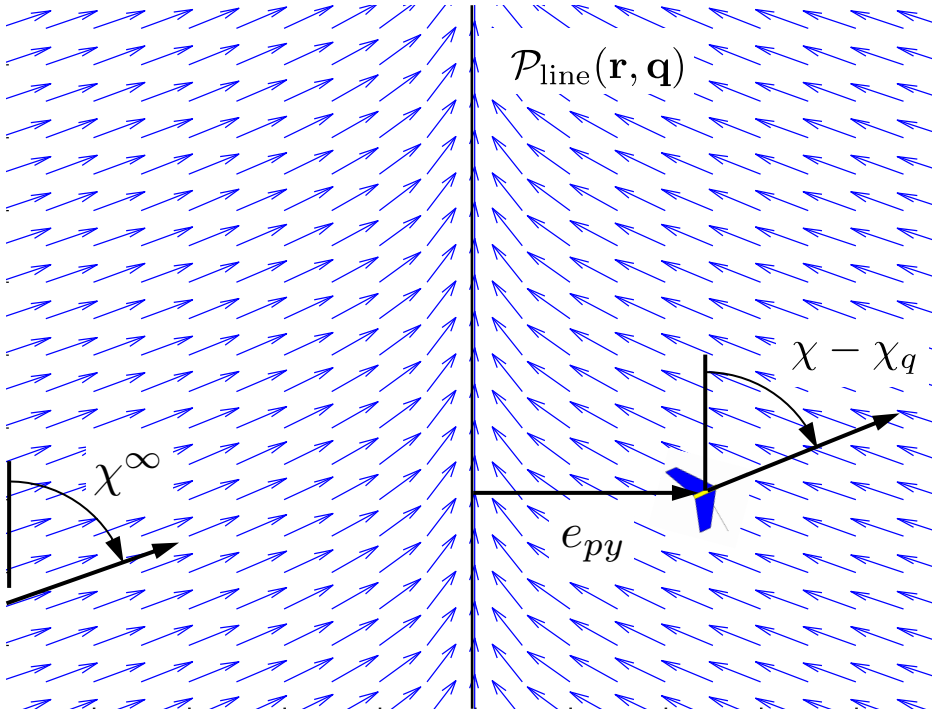
Regulate the cross-track error e_{py} to zero by commanding the course angle:

$$\begin{aligned} \dot{e}_{py} &= V_g \sin(\chi - \chi_q) \\ \ddot{\chi} &= b_{\dot{\chi}}(\dot{\chi}^c - \dot{\chi}) + b_{\chi}(\chi^c - \chi) \end{aligned}$$

Select χ^c so that $e_{py} \rightarrow 0$

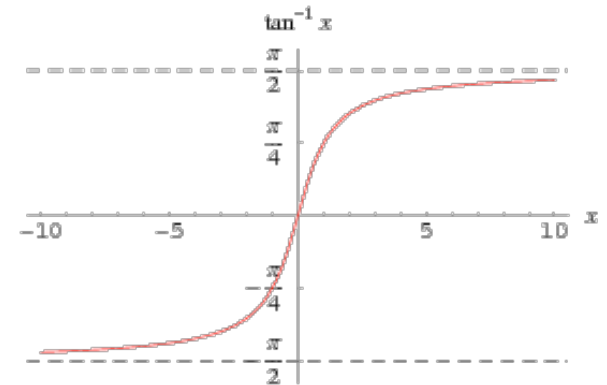


Lateral Tracking - Vector Field Concept



Desired course based on cross-track error:

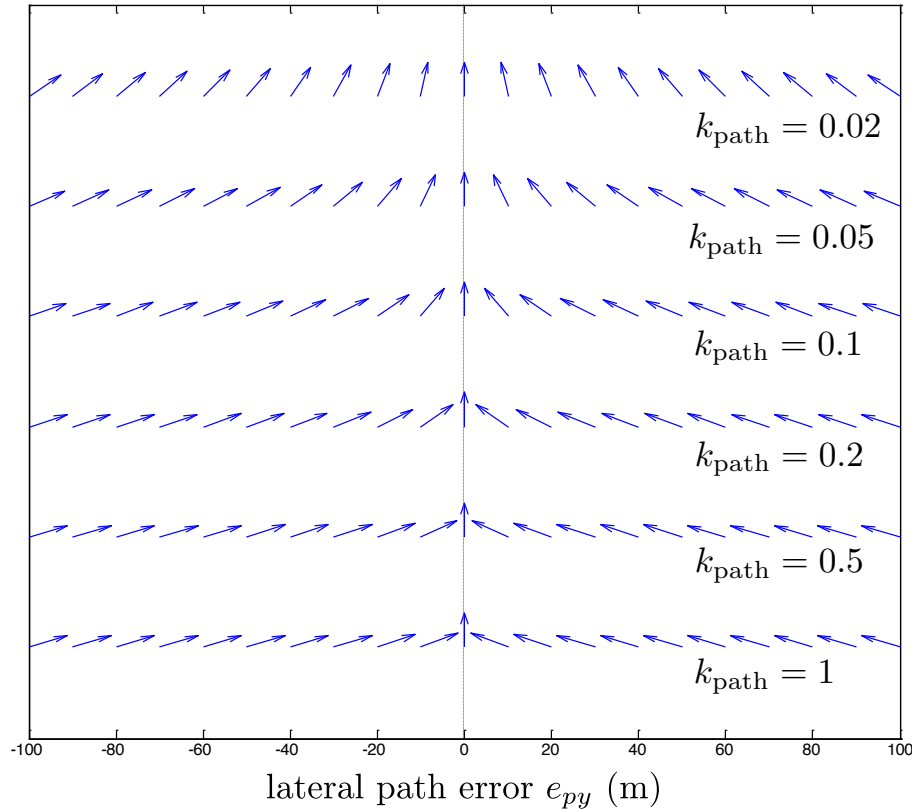
$$\chi_d(e_{py}) = -\chi^\infty \frac{2}{\pi} \tan^{-1}(k_{\text{path}} e_{py})$$



For a general line, the commanded course is

$$\chi^c = \chi_q - \chi^\infty \frac{2}{\pi} \tan^{-1}(k_{\text{path}} e_{py})$$

Vector Field Tuning



k_{path} is a positive constant that affects the rate of transition of the desired course

- k_{path} large $\rightarrow$ short, abrupt transition
- k_{path} small $\rightarrow$ long, gradual transition

Rule of thumb:

$$k_{\text{path}} \approx \frac{1}{R_{\min}},$$

where $R_{\min}$ is the minimum turn radius of the aircraft.

Lyapunov's 2nd Method

For a system having a state vector x , consider an energy-like (Lyapunov) function $V(x) : \mathbb{R}^n \mapsto \mathbb{R}$ such that

$$V(x) > 0, \forall x \neq 0 \text{ (positive definite)}$$

$$V(0) = 0$$

and

$$\dot{V}(x) < 0, \forall x \neq 0 \text{ (negative definite)}$$

$$\dot{V}(0) = 0.$$

If such a function $V(x)$ can be defined, then x goes to zero asymptotically and the system is stable.

Lateral Tracking Stability Analysis

Define the Lyapunov function $W(e_{py}) = \frac{1}{2}e_{py}^2$

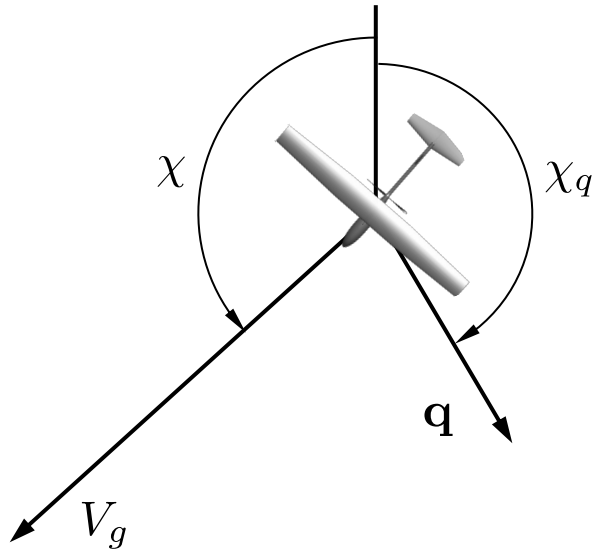
Assume that course controller works and $\chi = \chi_q + \chi^d(e_{py})$

Since

$$\begin{aligned}\dot{W} &= e_{py}\dot{e}_{py} \\ &= -V_a e_{py} \sin\left(\chi^\infty \frac{2}{\pi} \tan^{-1}(k_{\text{path}} e_{py})\right) \\ &< 0\end{aligned}$$

for $e_{py} \neq 0$, then $e_{py} \rightarrow 0$ asymptotically

Smallest Angle Turn Logic



$$\chi_q = \text{atan2}(q_e, q_n) + 2\pi m$$

$m \in \mathcal{N}$ is selected so that $-\pi \leq \chi_q - \chi \leq \pi$

Algorithm 3 Straight-line Following: $[h^c, \chi^c] = \text{followStraightLine}(\mathbf{r}, \mathbf{q}, \mathbf{p}, \chi)$

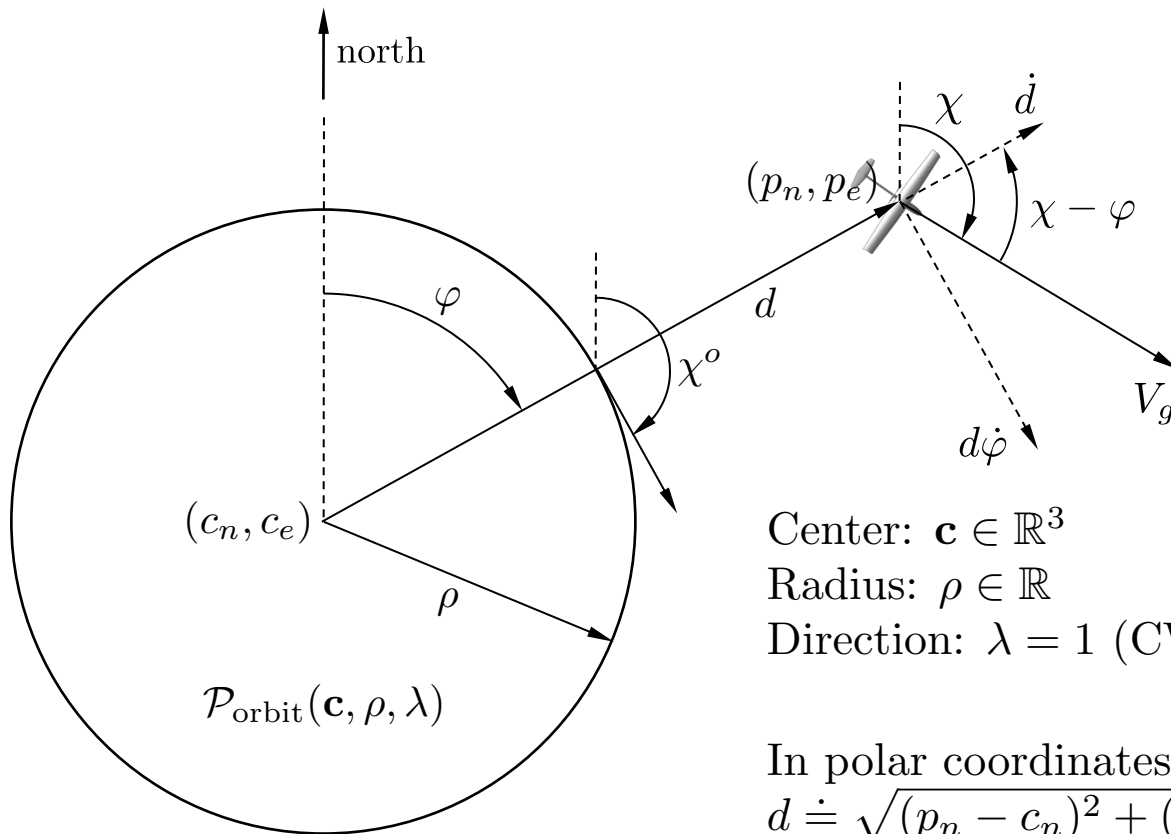
Input: Path definition $\mathbf{r} = (r_n, r_e, r_d)^\top$ and $\mathbf{q} = (q_n, q_e, q_d)^\top$, MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$, course χ , gains $\chi_\infty, k_{\text{path}}$, sample rate T_s .

- 1: Compute commanded altitude using equation (10.5).
 - 2: $\chi_q \leftarrow \text{atan2}(q_e, q_n)$
 - 3: **while** $\chi_q - \chi < -\pi$ **do**
 - 4: $\chi_q \leftarrow \chi_q + 2\pi$
 - 5: **end while**
 - 6: **while** $\chi_q - \chi > \pi$ **do**
 - 7: $\chi_q \leftarrow \chi_q - 2\pi$
 - 8: **end while**
 - 9: $e_{py} \leftarrow -\sin \chi_q (p_n - r_n) + \cos \chi_q (p_e - r_e)$
 - 10: Compute commanded course angle using equation (10.8).
 - 11: **return** h^c, χ^c
-

Orbit Following

Orbit definition:

$$\mathcal{P}_{\text{orbit}}(\mathbf{c}, \rho, \lambda) = \left\{ \mathbf{r} \in \mathbb{R}^3 : \mathbf{r} = \mathbf{c} + \lambda \rho \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \end{pmatrix}^\top, \varphi \in [0, 2\pi) \right\}$$



Center: $\mathbf{c} \in \mathbb{R}^3$

Radius: $\rho \in \mathbb{R}$

Direction: $\lambda = 1$ (CW) or $\lambda = -1$ (CCW)

In polar coordinates, the position of MAV given by:

$$d \doteq \sqrt{(p_n - c_n)^2 + (p_e - c_e)^2}:$$

$$\varphi \doteq \tan^{-1} \left(\frac{p_e - c_e}{p_n - c_n} \right):$$

Orbit Following

Easiest to analyze in polar coordinates.

Using

$$\begin{pmatrix} \dot{p}_n \\ \dot{p}_e \end{pmatrix} = \begin{pmatrix} V_g \cos \chi \\ V_g \sin \chi \end{pmatrix}$$

and converting to polar coordinates gives

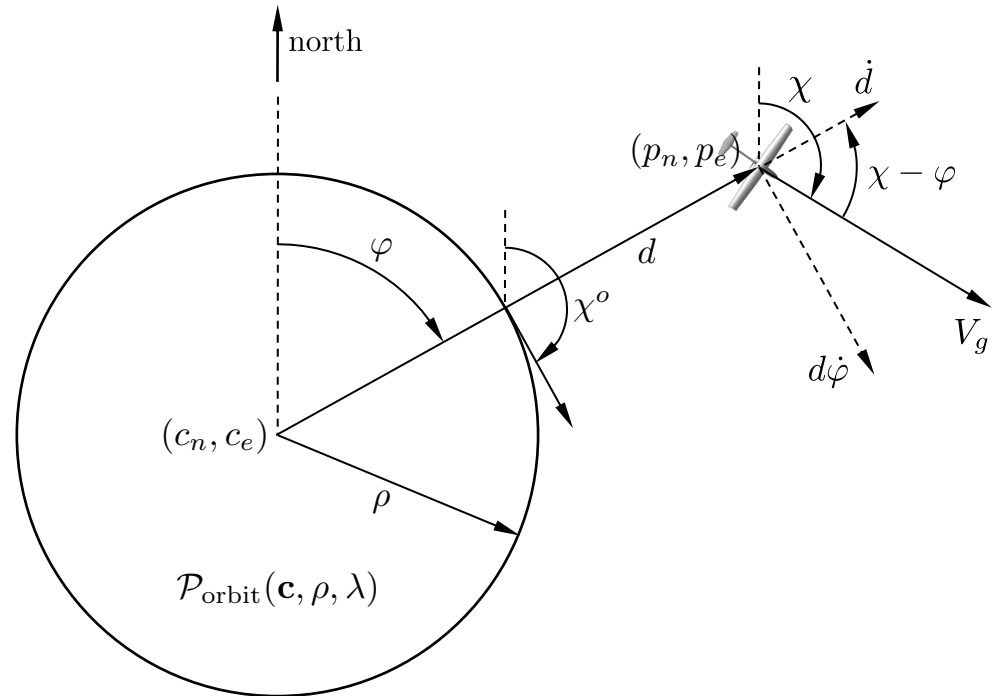
$$\begin{aligned} \begin{pmatrix} \dot{d} \\ d\dot{\varphi} \end{pmatrix} &= \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} \dot{p}_n \\ \dot{p}_e \end{pmatrix} \\ &= \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} V_g \cos \chi \\ V_g \sin \chi \end{pmatrix} \\ &= \begin{pmatrix} V_g \cos(\chi - \varphi) \\ V_g \sin(\chi - \varphi) \end{pmatrix} \end{aligned}$$

The result is

$$\dot{d} = V_g \cos(\chi - \varphi)$$

$$\dot{\varphi} = \frac{V_g}{d} \sin(\chi - \varphi)$$

$$\ddot{\chi} = -b_{\dot{\chi}}\dot{\chi} + b_{\chi}(\chi^c - \chi)$$



$$\dot{p} = R_z(\chi) \begin{bmatrix} V_g \\ 0 \end{bmatrix}$$

$$\dot{p}_T = R_z(\varphi) \begin{bmatrix} V_T \\ 0 \end{bmatrix}$$

$$d = R_z(-\varphi)(p - p_T)$$

$$\dot{d} = S(\varphi e_3)d + R_z(\chi - \varphi) \begin{bmatrix} V_g \\ 0 \end{bmatrix} + \begin{bmatrix} V_T \\ 0 \end{bmatrix}$$

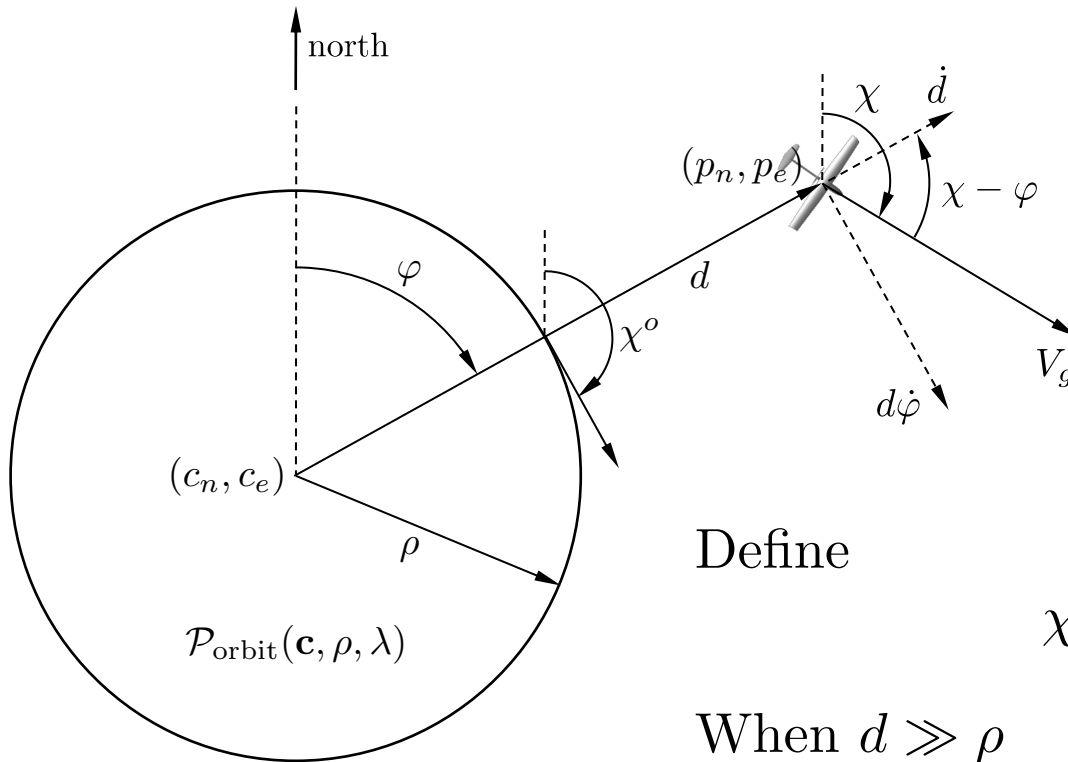
$$V = \frac{1}{2}d^T d$$

$$\dot{V} = d^T \dot{d} = d_x(\cos(\chi - \varphi)V_g - V_T) + d_y \sin(\chi - \varphi)$$

$$\begin{cases} \chi = \varphi + \delta(d_y) \\ V_T = \cos(\delta(d_y))V_g + \sigma(d_x) \end{cases}$$

$$\dot{V} = -d_x\sigma(d_x) - d_y\delta(d_y) < 0$$

Orbit Following



Define

$$\chi^o = \varphi + \lambda \frac{\pi}{2}$$

When $d \gg \rho \rightarrow \chi_d \approx \chi^o + \lambda \frac{\pi}{2}$.

When $d = \rho \rightarrow \chi_d = \chi^o$

Therefore, let the desired course angle be

$$\chi_d(d-\rho, \lambda) = \chi^o + \lambda \tan^{-1} \left(k_{\text{orbit}} \left(\frac{d-\rho}{\rho} \right) \right)$$

Orbit Tracking Stability Analysis

Define the Lyapunov function $W = \frac{1}{2}(d - \rho)^2$

Assume that course controller works and $\chi = \chi^d(d - \rho, \lambda)$

Since

$$\begin{aligned}\dot{W} &= (d - \rho)\dot{d} \\ &= (d - \rho)(V_g \cos(\chi - \varphi)) \\ &= -V_g(d - \rho) \sin \left(\tan^{-1} \left(k_{\text{orbit}} \left(\frac{d - \rho}{\rho} \right) \right) \right) \\ &< 0\end{aligned}$$

for $d - \rho \neq 0$, then $d - \rho \rightarrow 0$ asymptotically

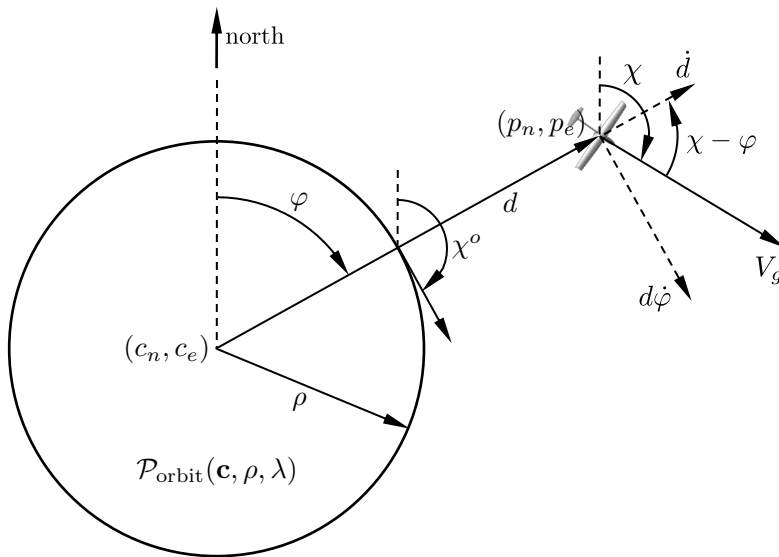
Orbit Following

The commanded course is

$$\chi^c(t) = \varphi + \lambda \left[\frac{\pi}{2} + \tan^{-1} \left(k_{\text{orbit}} \left(\frac{d - \rho}{\rho} \right) \right) \right]$$

The orbit angle must be wrapped:

$$\varphi = \text{atan2}(p_e - c_e, p_n - c_n) + 2\pi m$$



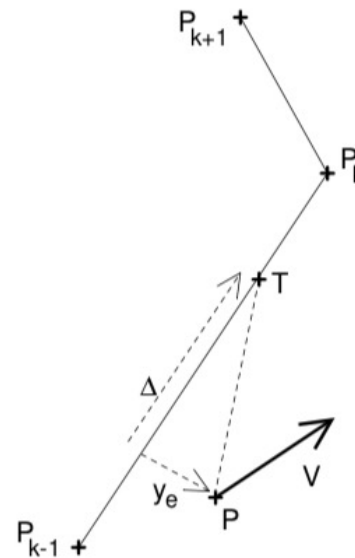
Algorithm 4 Circular Orbit Following: $[h^c, \chi^c] = \text{followOrbit}(\mathbf{c}, \rho, \lambda, \mathbf{p}, \chi)$

Input: Orbit center $\mathbf{c} = (c_n, c_e, c_d)^\top$, radius ρ , and direction λ , MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$, course χ , gains k_{orbit} , sample rate T_s .

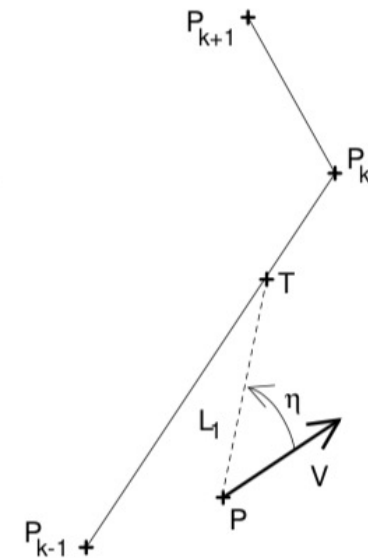
- 1: $h^c \leftarrow -c_d$
 - 2: $d \leftarrow \sqrt{(p_n - c_n)^2 + (p_e - c_e)^2}$
 - 3: $\varphi \leftarrow \text{atan2}(p_e - c_e, p_n - c_n)$
 - 4: **while** $\varphi - \chi < -\pi$ **do**
 - 5: $\varphi \leftarrow \varphi + 2\pi$
 - 6: **end while**
 - 7: **while** $\varphi - \chi > \pi$ **do**
 - 8: $\varphi \leftarrow \varphi - 2\pi$
 - 9: **end while**
 - 10: Compute commanded course angle using equation (10.13).
 - 11: **return** h^c, χ^c
-

Lookahead distance

Errors may be computed with
a **lookahead distance**:
2 popular guidance approaches



LOS



L1

- Line-of-sight (LOS)
- L1 guidance law
Adequate for straight or circular segments